Project Workflow: End-to-End Analytical Pipeline

Phase 3: Operational Pipeline & Step Execution

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1. Operational Pipeline Execution Steps

The analytical lifecycle proceeds through five deterministic, automated phases to transform unstructured sports history into validated business intelligence.

- Data Ingestion: Load raw Olympic records and join NOC region metadata.
- Data Cleaning: Impute missing attributes, filter modern era (1980-2016), and standardize medal categories.
- Aggregation: Group by [games_id, noc] and compute G, S, B, total medals, and weighted scores.
- Visualization: Author interactive views in Tableau and publish to Tableau Public.
- Verification: Execute automated test suite to verify 100% data and mathematical integrity.